L1–L2 Differences in Linguistic and Perceptual Integration

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Embodied simulation theories propose that language comprehension involves reactivating perceptual and motor representations grounded in prior sensorimotor experience (Barsalou, 1999, 2008). A central question concerns whether, and how, perceptual information is recruited during online sentence processing. The sentence–picture verification (SPV) paradigm has been widely used to address this question: after reading a sentence, speakers judge whether a subsequent picture depicts a mentioned object, and responses are typically faster when the picture matches perceptual features implied—but not explicitly stated—by the sentence (Zwaan & Madden, 2005). Although this congruency advantage is often interpreted as evidence for perceptual simulation during comprehension, it remains unclear whether it reflects purely perceptual representations, purely linguistic context, or an interaction of the two. In addition, a related debate concerns L2 comprehension, where congruency effects in the SPV task are often reduced or absent (e.g., Norman & Peleg, 2022). It remains unclear whether such attenuated effects reflect weaker or less automatic perceptual grounding in L2, or whether they instead arise from differences in language experience that modulate how linguistic context constrains access to perceptual representations.

In the present study, we adapted the SPV paradigm by orthogonally manipulating linguistic distributional cue strength and picture typicality (see Table 1). Linguistic distributional cues (strong vs. weak) were defined in terms of corpus-derived co-occurrence patterns, indexing how strongly a sentence licenses a particular event based on speakers' accumulated linguistic experience, rather than on situational or pragmatic typicality. Picture typicality captured the perceptual prototypicality of object forms (Palmer et al., 1981), contrasting canonical exemplars (e.g., a whole lemon) with less typical variants (e.g., a lemon slice). This design allowed us to test whether the classic SPV congruency effect arises from perceptual representations alone or from their interaction with linguistic context. Eighty native speakers of Chinese (mean age = 20.1, 28 males) and eighty advanced L2 learners of Chinese (mean age = 21.9, 17 males; university-level) completed the experiment. L2 participants met advanced-level thresholds on vocabulary (Pelzl et al., 2019) and communicative ability measures (Bachman & Palmer, 1989), allowing us to examine whether language experience modulates the interaction between linguistic cues and perceptual representations. Reaction times were analyzed using linear mixed-effects models with language group, cue strength, and picture typicality as fixed effects.

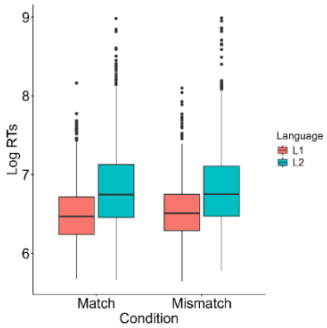
Results revealed that L2 speakers responded more slowly than L1 speakers ($\beta = 0.331$, $SE = 0.050$, $p < .001$). Responses were slower following weak than strong linguistic cues ($\beta = 0.048$, $SE = 0.022$, $p = .027$) and for atypical than typical pictures ($\beta = 0.148$, $SE = 0.022$, $p < .001$). Critically, cue strength interacted with picture typicality ($\beta = -0.069$, $SE = 0.031$, $p = .028$), demonstrating that typical pictures were recognized faster than atypical ones under strong cues, with a reduced typicality advantage under weak cues. Cue strength also interacted with language ($\beta = -0.070$, $SE = 0.032$, $p = .028$): L1 speakers responded more slowly under weak than strong

cues, whereas this cue effect was attenuated in L2 speakers.

These findings suggest that congruency effects arise from interactions between linguistic context and perceptual representations, rather than being driven solely by perceptual simulation. Although L1 and L2 speakers differed in their sensitivity to linguistic cue strength, both groups exhibited similar patterns of picture typicality effects, indicating that reduced congruency effects in L2 are not necessarily due to weaker perceptual grounding per se, but instead reflect differences in how linguistic experience shapes the use of distributional cues during comprehension.

Table 1. Sample stimuli used in the current study.

Sentence	Picture	Sentence Cue Strength	Picture Typicality	Condition	Response
他看到树上有苹果。 He saw apple on the tree.		Strong	Typical	Match	Yes
他看到沙拉里有苹果。 He saw apple in the salad.		Strong	Atypical	Mismatch	Yes
他看到沙拉里有苹果。 He saw apple in the salad.		Weak	Typical	Match	Yes
他看到树上有苹果。 He saw apple on the tree.		Weak	Atypical	Mismatch	Yes



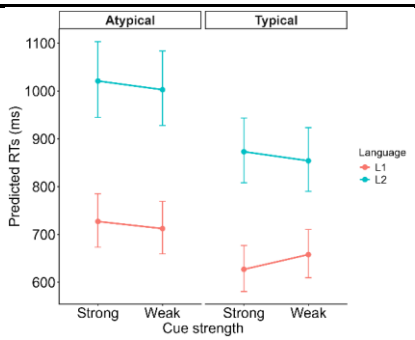


Figure 1. Log RT Distributions Across Conditions and Languages. **Figure 2.** Predicted RTs from the model.

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